4/4/2026

Security Assessment of an Unknown System in an Isolated Virtual Network

Course: Network Security

Section: A

Submitted to: Rifat Tasnim Anannya

Target VM Used: marlinspike

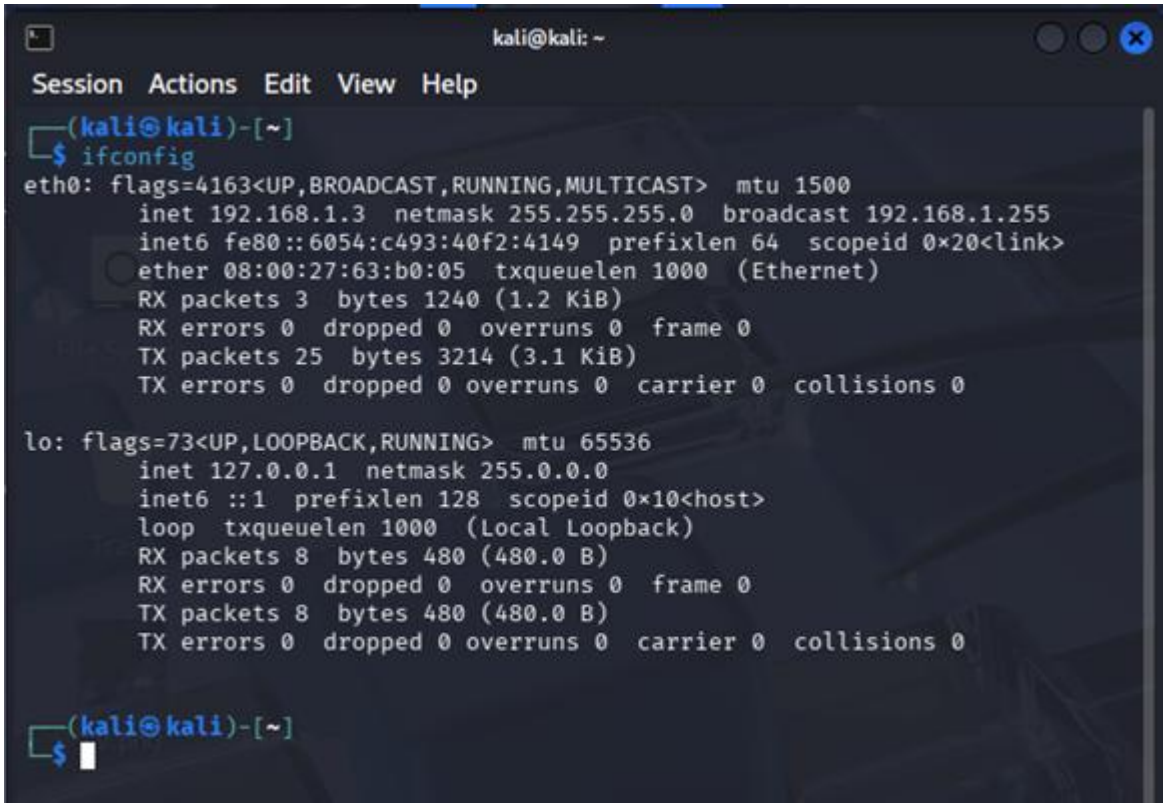
NAT Network Used: Yes

Name: SHOWRAV GHOSH

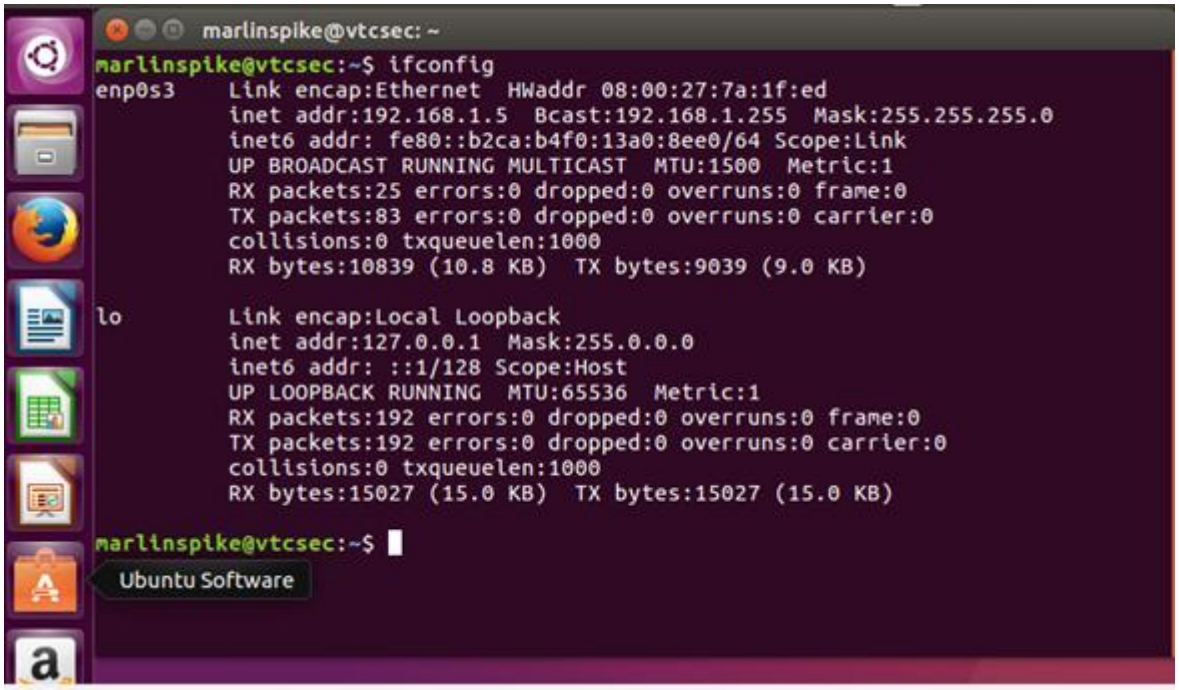
ID: 23-50666-1

IP Address

Kali ip address (Attacker IP):



Marlinspike ip address (Target Ip):



NMAP

Service & Version Detection:

```
kali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 09:43 EDT
Nmap scan report for 192.168.1.5
Host is up (0.00041s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     ProFTPD 1.3.3c
22/tcp open  ssh     OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol
2.0)
80/tcp open  http    Apache httpd 2.4.18 ((Ubuntu))
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.77 seconds
```

OS detection:

```
kali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap -O 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 09:44 EDT
Nmap scan report for 192.168.1.5
Host is up (0.0015s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.65 seconds
```

Aggressive Scan (All in One):

```
Session  Actions  Edit  View  Help
└─$ nmap -A 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 09:46 EDT
Nmap scan report for 192.168.1.5
Host is up (0.0012s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     ProFTPD 1.3.3c
22/tcp open  ssh     OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol
2.0)
| ssh-hostkey:
|   2048 d6:01:90:39:2d:8f:46:fb:03:86:73:b3:3c:54:7e:54 (RSA)
|   256 f1:f3:c0:dd:ba:a4:85:f7:13:9a:da:3a:bb:4d:93:04 (ECDSA)
|_  256 12:e2:98:d2:a3:e7:36:4f:be:6b:ce:36:6b:7e:0d:9e (ED25519)
80/tcp open  http    Apache httpd 2.4.18 ((Ubuntu))
|_http-title: Site doesn't have a title (text/html).
|_http-server-header: Apache/2.4.18 (Ubuntu)
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT     ADDRESS
1   1.17 ms 192.168.1.5

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.17 seconds
```

Vulnerability Script Scan:

```
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap --script vuln 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 09:59 EDT
Nmap scan report for 192.168.1.5
Host is up (0.00039s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
| ftp-proftpd-backdoor:
|   This installation has been backdoored.
|   Command: id
|_  Results: uid=0(root) gid=0(root) groups=0(root),65534(nogroup)
22/tcp open  ssh
80/tcp open  http
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
| http-enum:
|_  /secret/: Potentially interesting folder
| http-slowloris-check:
|   VULNERABLE:
|   Slowloris DOS attack
|     State: LIKELY VULNERABLE
|     IDs:  CVE:CVE-2007-6750
|       Slowloris tries to keep many connections to the target web server ope
n and hold
|       them open as long as possible.  It accomplishes this by opening conne
ctions to
|       the target web server and sending a partial request. By doing so, it
starves
|       the http server's resources causing Denial Of Service.
|
|     Disclosure date: 2009-09-17
|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|_      http://ha.ckers.org/slowloris/
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 322.01 seconds
```

Ping Scan (Check host is alive):

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 10:06 EDT
Nmap scan report for 192.168.1.5
Host is up (0.00076s latency).
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
```

Scan top 1000 ports (default):

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 10:07 EDT
Nmap scan report for 192.168.1.5
Host is up (0.00038s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 0.59 seconds
```

Scan specific ports:

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ nmap -p 21,22,80 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 10:08 EDT
Nmap scan report for 192.168.1.5
Host is up (0.0015s latency).

PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 4.16 seconds
```

Scan a port range:

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ nmap -p 1-1000 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 10:10 EDT
Nmap scan report for 192.168.1.5
Host is up (0.00034s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 0.38 seconds
```

Scan with default script (Defect Vulnerabilities):

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ nmap -sC 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 10:10 EDT
Nmap scan report for 192.168.1.5
Host is up (0.00034s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
| ssh-hostkey:
|   2048 d6:01:90:39:2d:8f:46:fb:03:86:73:b3:3c:54:7e:54 (RSA)
|   256 f1:f3:c0:dd:ba:a4:85:f7:13:9a:da:3a:bb:4d:93:04 (ECDSA)
|_  256 12:e2:98:d2:a3:e7:36:4f:be:6b:ce:36:6b:7e:0d:9e (ED25519)
80/tcp open  http
|_http-title: Site doesn't have a title (text/html).
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 1.01 seconds
```

Stealth scan:

```
Session  Actions  Edit  View  Help
└─$ nmap -sS 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 10:11 EDT
Nmap scan report for 192.168.1.5
Host is up (0.00035s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds
```

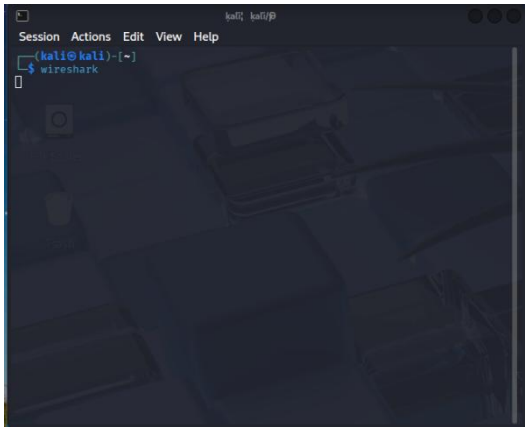
UDP port scan (Check DNS, DHCP services):

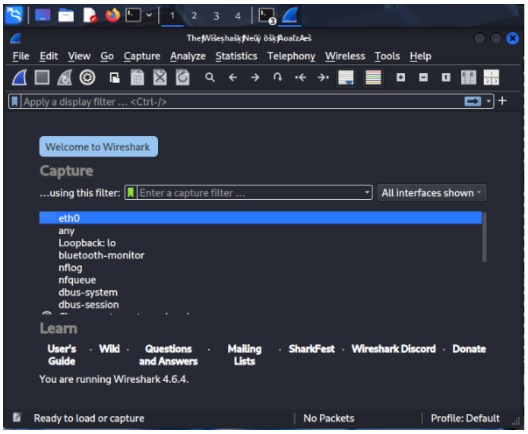
```
┌──(kali㉿kali)-[~]
└─$ nmap -sU 192.168.1.5
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-02 10:12 EDT
Stats: 0:07:22 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 49.23% done; ETC: 10:27 (0:07:36 remaining)
Stats: 0:10:47 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 68.90% done; ETC: 10:27 (0:04:52 remaining)
Stats: 0:13:41 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 85.52% done; ETC: 10:28 (0:02:19 remaining)
Nmap scan report for 192.168.1.5
Host is up (0.0014s latency).
All 1000 scanned ports on 192.168.1.5 are in ignored states.
Not shown: 957 closed udp ports (port-unreach), 43 open|filtered udp ports (n
o-response)
MAC Address: 08:00:27:7A:1F:ED (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 1035.93 seconds
```

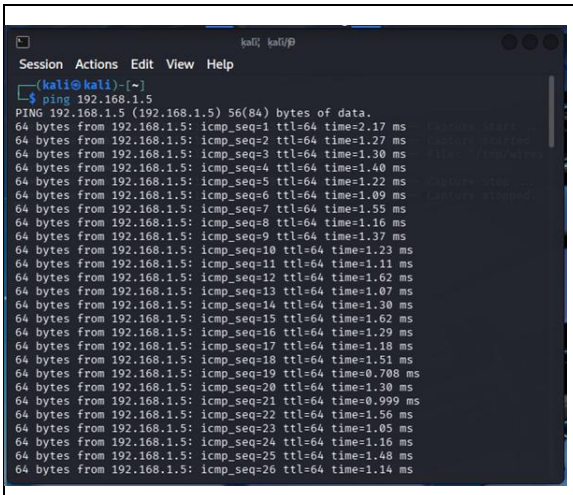
WIRESHARK

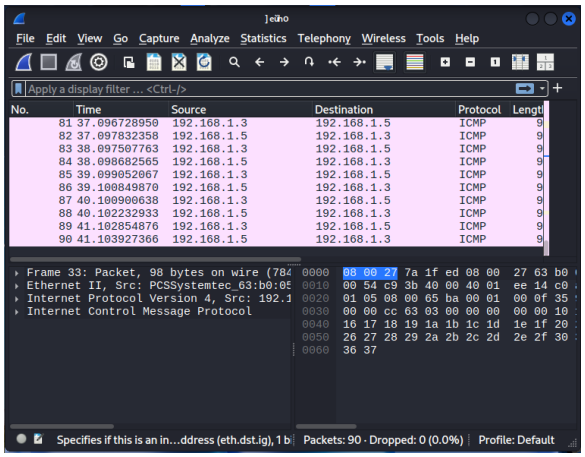
Start Wireshark:



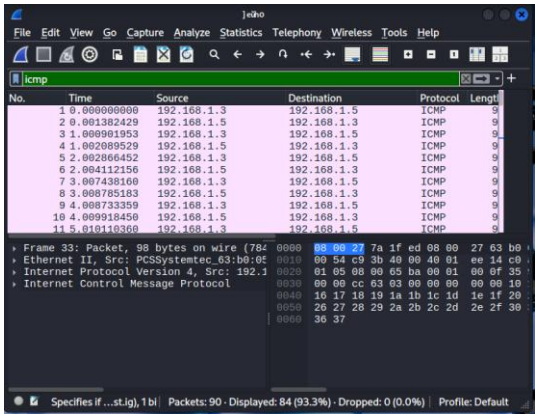


Generate Traffic (ICMP):

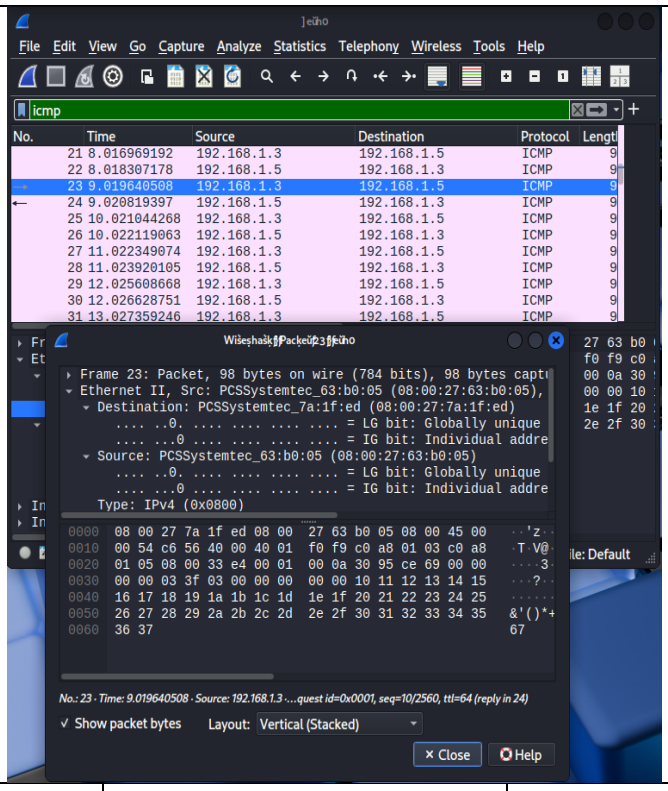


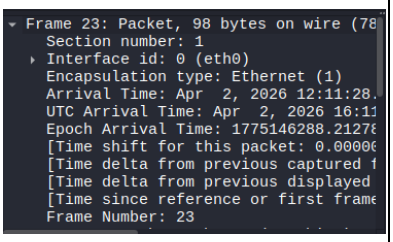


Filter ICMP packets:

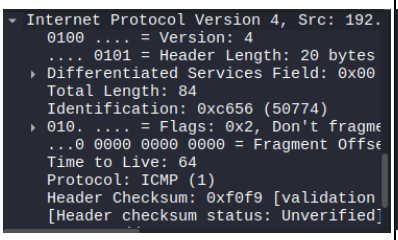


Analyse a packet:

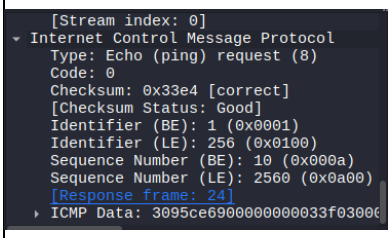




Expand protocol layer

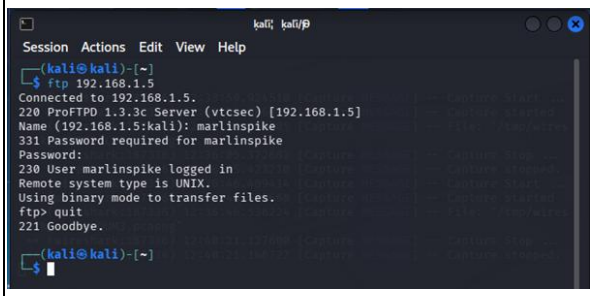


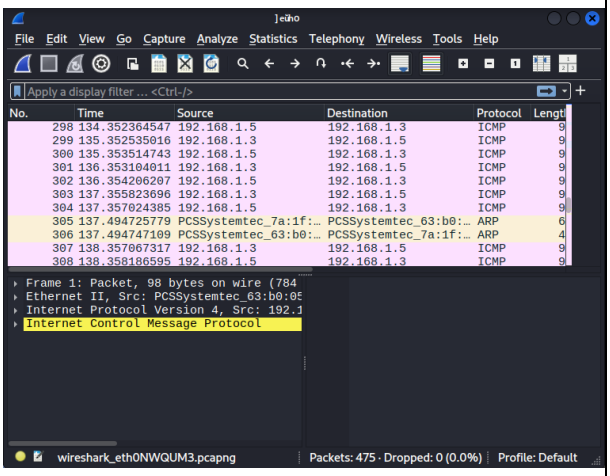
Expand protocol layer



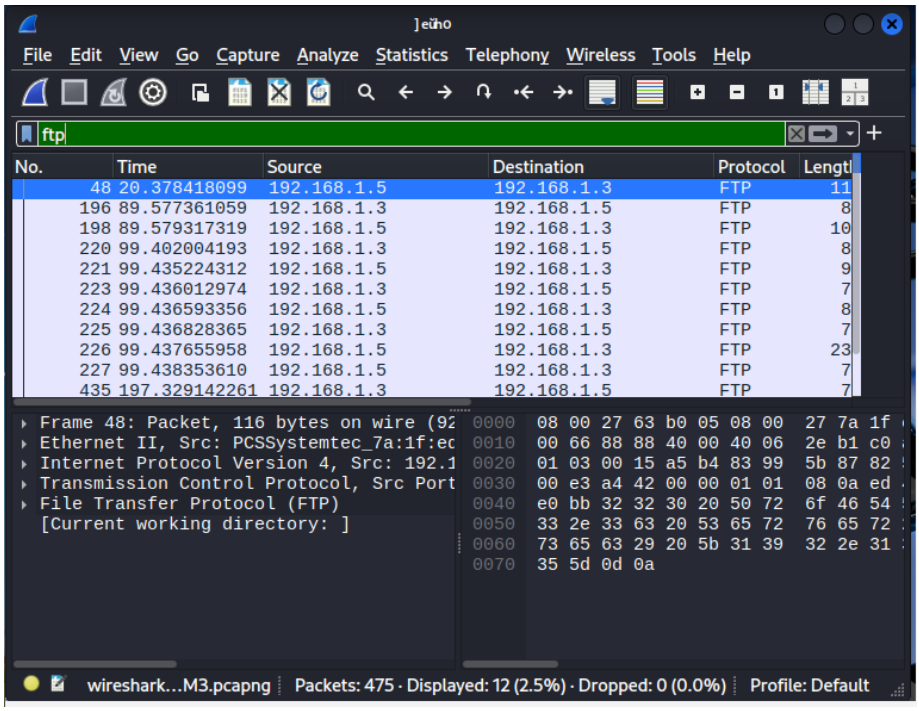
Expand protocol layer

Generate Login traffic (FTP):

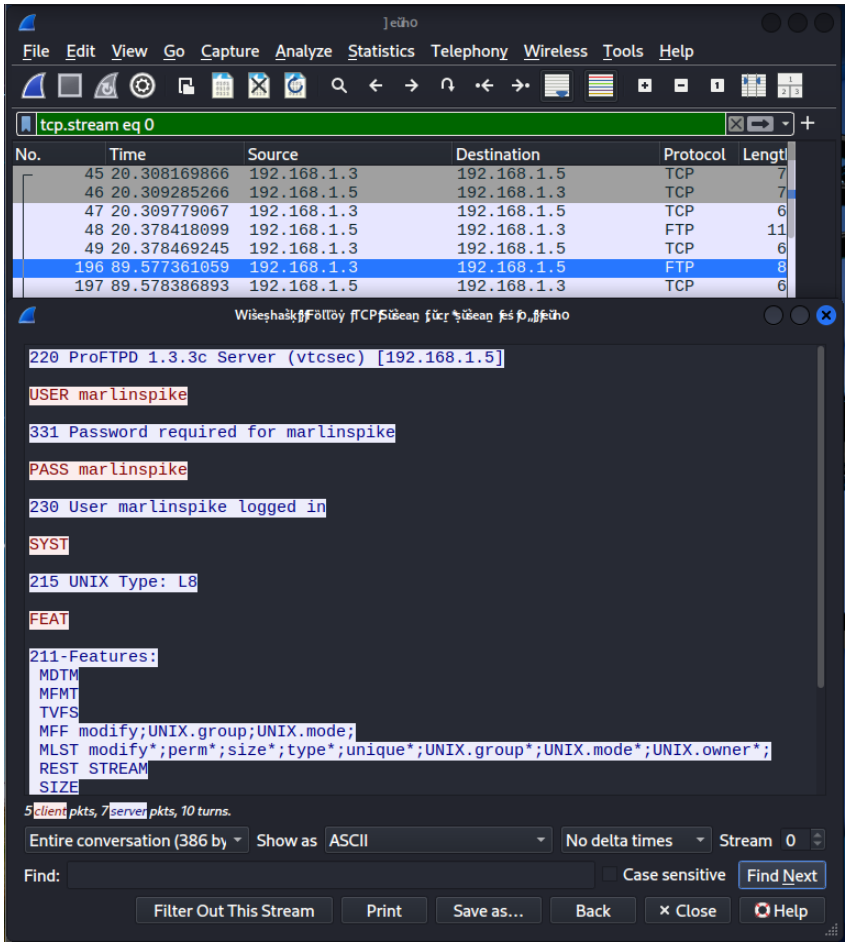




Filter FTP Packets:



Follow TCP Stream:



John The Ripper

Find Target ip:

My kali ip:

```
root@vtcsec: /home/marlinspike
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.3  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::6054:c493:40f2:4149  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:63:b0:05  txqueuelen 1000  (Ethernet)
        RX packets 44  bytes 7365 (7.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 558  bytes 35792 (34.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Scan network to locate Marlinspike:

```
root@vtcsec: /home/marlinspike
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-03 00:44 -0400
Nmap scan report for 192.168.1.1
Host is up (0.00067s latency).
MAC Address: 52:55:C0:A8:01:01 (Unknown)
Nmap scan report for 192.168.1.2
Host is up (0.00030s latency).
MAC Address: 08:00:27:32:B8:D4 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.5
Host is up (0.00091s latency).
MAC Address: 08:00:27:7A:1F:ED (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.3
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 8.53 seconds
```

Connect to Marlinspike Using SSH:

```
root@vtcsec: /home/marlinspike
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ssh marlinspike@192.168.1.5
The authenticity of host '192.168.1.5 (192.168.1.5)' can't be established.
ED25519 key fingerprint is: SHA256:ZEGvF8tQ4SMYJOaKofsm1TFy5G+/ey3R7Fxd9X4eQo
Q
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.5' (ED25519) to the list of known hosts
.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
marlinspike@192.168.1.5's password:
Welcome to Ubuntu 16.04.3 LTS (GNU/Linux 4.10.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

646 packages can be updated.
500 updates are security updates.

New release '18.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Become Root User on Marlinspike:

```
kali@kali: ~
Session  Actions  Edit  View  Help
marlinspike@vtcsec:~$ sudo su
[sudo] password for marlinspike:
root@vtcsec:/home/marlinspike# whoami
root
```

Copy Password Files and Fix Permissions(On Marlinspike as root – via Kali SSH terminal):

```
root@vtcsec:/home/marlinspike# whoami
root
root@vtcsec:/home/marlinspike# cp /etc/passwd /tmp/passwd
root@vtcsec:/home/marlinspike# cp /etc/shadow /tmp/shadow
root@vtcsec:/home/marlinspike# chmod 644 /tmp/passwd
root@vtcsec:/home/marlinspike# chmod 644 /tmp/shadow
```

Exit Marlinspike Session:

```
root@vtcsec:/home/marlinspike# exit
exit
marlinspike@vtcsec:~$ exit
logout
Connection to 192.168.1.5 closed.

┌──(kali㉿kali)-[~]
```

Copy Password Files from Marlinspike to Kali (On Kali Linux terminal):

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ scp -oHostKeyAlgorithms=+ssh-rsa -oPubkeyAcceptedAlgorithms=+ssh-rsa marl
inspike@192.168.1.5:/tmp/passwd .
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
marlinspike@192.168.1.5's password:
passwd                                    100% 2364   397.1KB/s   00:00

┌──(kali㉿kali)-[~]
└─$ scp -oHostKeyAlgorithms=+ssh-rsa -oPubkeyAcceptedAlgorithms=+ssh-rsa marl
inspike@192.168.1.5:/tmp/shadow .
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
marlinspike@192.168.1.5's password:
shadow                                    100% 1305   321.0KB/s   00:00
```

Verify File Transfer:

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ls
Desktop    Downloads  passwd    Public   Templates
Documents  Music      Pictures  shadow   Videos
```

Combine Files for John the Ripper (On Kali terminal):

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ unshadow passwd shadow > crackme.txt
Created directory: /home/kali/.john
```

Start Password Cracking (On Kali terminal):

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ john crackme.txt
Using default input encoding: UTF-8
Loaded 1 password hash (sha512crypt, crypt(3) $6$ [SHA512 256/256 AVX2 4x])
Cost 1 (iteration count) is 5000 for all loaded hashes
Will run 2 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 5 candidates buffered for the current salt, minimum 8 needed fo
r performance.
marlinspike      (marlinspike)
1g 0:00:00:00 DONE 1/3 (2026-04-03 01:05) 16.66g/s 83.33p/s 83.33c/s 83.33C/s
 marlinspike..marli
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

Show Cracked Passwords (On Kali terminal):

```
┌──(kali㉿kali)-[~]
└─$ john --show crackme.txt
marlinspike:marlinspike:1000:1000:marlinspike,,,:/home/marlinspike:/bin/bash

1 password hash cracked, 0 left
```